Year 2 Assessed Problems

Semester 2

Assessed Problems 2

SOLUTIONS TO BE SUBMITTED ON CANVAS BY

Wednesday 11th February

at 17:00

Y2 Electromagnetism 2
Assessed problem sheet 1 (weeks 1–2)

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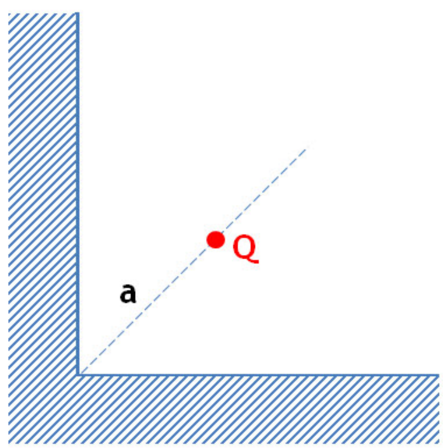
1) Three point electrical charges of equal sign and magnitude are placed at the vertices of an equilateral triangle. What is the magnitude of the charge that should be placed at the centre of the triangle to put the system in equilibrium? Would the equilibrium be stable? [**3**]

2) Find the electrostatic potential at the centre of a uniformly charged solid sphere of radius R with a total electric charge Q. [**3**]

3) A point charge Q is located on the bisector of a right dihedral angle formed by two grounded semi-infinite grounded conductive plates, at the distance a from the edge of the dihedral angle. Identify the system of image charges which describes the electric field inside the angle. Prove that the boundary conditions are satisfied by the system of charges. Find the direction and magnitude of the force acting on the charge Q due to the induced charges. [**4**]



Y2 Neutrino Physics
Assessed problem sheet 1 (weeks 1–2)

The proposed LHeC collider at CERN would bring beams of positrons (e^+) and protons (p) with momenta of 100 GeV/c and 7 TeV/c, respectively, into head-on collisions.

a) Calculate the proton velocity β, quoting your result as $(1 - \beta)$. [**1**]

b) Calculate the energy E_{CM} available in the positron-proton centre of mass frame. Provide your answer in GeV. [**1**]

c) Estimate the order of magnitude of the distance scale that can be probed using the above energy E_{CM}. [**1**]

d) Consider a positron-proton collision experiment with an e^+ beam incident on a stationary proton target. Which positron energy is required to achieve the LHeC energy in the e^+p centre of mass frame? Compare it with the beam energies provided by modern accelerators. [**2**]

Optics Assessed Problem sheet 1

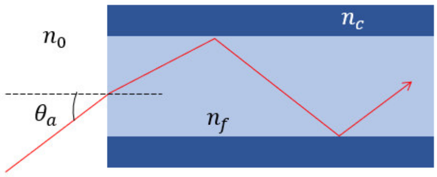
1 (a) i. Please use Fermat's Principle to derive the law of refraction, i.e. [3]

$$n_i \sin\theta_i = n_t \sin\theta_t$$

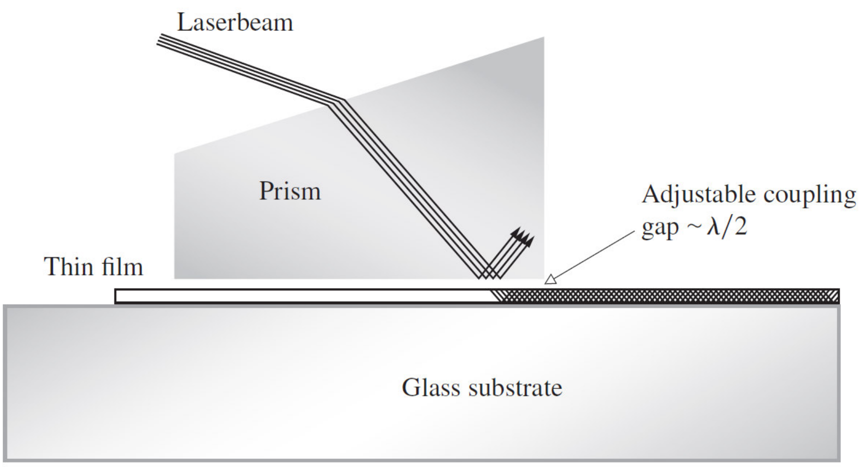
ii. A beam of infrared light (wavelength in free space = 10.6 μm) is incident at 40^0 on the surface of germanium which is transparent for those waves and has a refractive index of 4. What is the wavelength of light beam in the germanium? Calculate the transmission angle. [2]

(b) i. The acceptance angle of a multi-mode fibre: [3]
The acceptance angle θ_a of an optical fibre is defined as the maximum angle for the incident light that can propagate through a fibre.

Assuming there is a step-index multi-mode fibre as illustrated in the figure, can you find an expression for θ_a of such a fibre?



ii. Figure below shows a prism-coupler arrangement developed at the Bell Telephone Laboratories. Its function is to feed a laser beam into a thin (0.00001 inch) transparent film, which then serves as a sort of waveguide. One application is that of thin-film laser-beam circuitry -a kind of integrated optics. How do you think it works? [2]



Year 2: Observational Astronomy

Assessed Problem Sheet #1

1. A star cluster contains 357 identical stars, each of apparent magnitude, $m = 15$, at $550\mathrm{nm}$. Calculate the apparent magnitude of the star cluster as a whole.
   [1 mark]

2. The dark adapted human eye (i.e. fully dilated pupil) can see objects as faint as 6th magnitude. Is the star cluster detectable by the dark adapted human eye? Justify your answer.
   [1 mark]

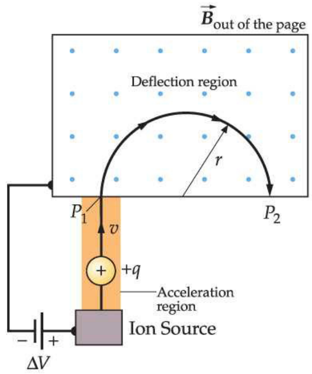
3. The celestial coordinates of the Andromeda Galaxy, M31, are $(\alpha, \delta) = (00h\,42m\,44.3s, +41d\,16m\,9s)$. Convert these coordinates in to units of decimal degrees. Give your answers to 5 decimal places.
   [2 marks]

4. At what time (to the nearest hour) will M31 transit the meridian on July 21st and October 21st? On which of these two nights would you prefer to observe M31, and why?
   [3 marks]

5. Rank the following locations (that are home to powerful telescopes) in order of preference for observing M31: Mauna Kea $(\lambda, \beta) = (-155.3°, +19.8°)$, Cerro Paranal $(\lambda, \beta) = (-70.4°, -24.6°)$, Siding Spring $(\lambda, \beta) = (+149.1°, -31.3°)$, La Palma $(\lambda, \beta) = (-17.8°, +28.7°)$, where λ and β are the longitude and latitude of each site respectively. Justify your ranking.
   [2 marks]

6. How would your answer to the previous question change if a supernova exploded in M31 at 12noon GMT on the relevant day, and it was essential to observe as soon as possible?
   [1 mark]

1. A mass spectrometer accelerates nuclei in an electric field. They then enter a magnetic field of $0.5\,\mathrm{T}$ perpendicular to the flight path. After a semicircle, they hit a screen where they are measured. For this exercise, the mass spectrometer is used to separate ${}^{14}_{6}\mathrm{C}$ and ${}^{12}_{6}\mathrm{C}$. These have a mass of $m_{^{14}\mathrm{C}} = 14.0032\,\mathrm{u}$ and $m_{^{14}\mathrm{C}} = 12\,\mathrm{u}$ (atomic masses).



   (a) The nuclei are accelerated to a kinetic energy of $E_{kin} = 60\,\mathrm{MeV}$. Calculate their momenta. (Hint: $E_{kin} = E - E_0$ with $E_0 = mc^2$) **[1]**

   (b) The nuclei are accelerated in an electric field. What voltage difference is required to reach this energy? You can assume that the nuclei are completely stripped of their electrons. **[1]**

   (c) What is the distance between the points where ${}^{14}_{6}\mathrm{C}$ and ${}^{12}_{6}\mathrm{C}$ are measured on the screen? **[1]**

   (d) Calculate the binding energy of ${}^{14}_{6}\mathrm{C}$. **[1]**

   (e) Calculate the energy released when ${}^{14}_{6}\mathrm{C}$ decays to ${}^{14}_{7}\mathrm{N}$ via β^- decay. The binding energy of ${}^{14}_{7}\mathrm{N}$ is $B = 104.658\,\mathrm{MeV}$ **[1]**

Hints:
$1\,\mathrm{u} = 931.494\,\mathrm{MeV}/c^2$
$m_{^{1}\mathrm{H}} = 1.007825\,\mathrm{u}$